

ABHAY SAHARAWAT

Roorkee, Uttarakhand (Open to Relocate)

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SUMMARY

Enthusiastic undergraduate specializing in AI Data Science, with hands-on project and research experience, seeking an Analyst Trainee role to apply strong analytical skills, technical expertise, and effective communication to solve complex problems and deliver meaningful, data-driven insights.

TECHNICAL SKILLS

- **Languages:** C, Python, Java, HTML, CSS.
- **Libraries Frameworks:** Numpy, Pandas, Matplotlib, Scikit-Learn, OpenCV, YOLO.
- **CS Concepts:** Object-Oriented Programming, DBMS, Operating System, Computer Networks, Cloud Computing.
- **Databases Query Languages:** SQL: SQL.
- **Tools:** Microsoft Excel, Microsoft PowerPoint, Microsoft Word.
- **Domain Skills:** Machine Learning, Data Analysis, Computer Vision, Real-Time Systems, Heuristic Analysis, Entropy Analysis, Cryptography, GUI Development (Tkinter), Data Visualization.
- **Soft Skills:** Analytical Thinking, Problem Solving, Effective Communication, Attention to Detail, Team Collaboration, Time Management, Adaptability, Quick Learner.

PROJECTS

Real-Time Traffic Analysis System | Python, OpenCV, YOLO May 2026

- Developed a real-time traffic analysis system using YOLO and OpenCV for vehicle detection and lane-wise traffic monitoring.
- Implemented vehicle counting and traffic intensity classification (Smooth/Heavy) for multiple lanes.
- Designed region-based lane detection with live annotations, bounding boxes, and traffic statistics overlay.
- Processed video streams in real time for intelligent traffic surveillance and analysis.

PassGenX – Password Generator with Entropy-Based Strength Checker Oct 2025

- Developed a secure password generation system using cryptographically strong random functions to eliminate predictable patterns.
- Designed a password strength analyzer based on entropy and resistance to brute-force attacks.
- Implemented optional local encryption for securely storing user credentials.

Lightweight Malware Scanner – Python-Based Detection System | Python, Tkinter, hashlib Nov 2024

- Built a malware detection system using SHA-1 hashing and heuristic analysis to identify suspicious files.
- Developed a Tkinter-based GUI for real-time file scanning and result visualization.
- Implemented a quarantine mechanism to isolate infected files securely.
- Enabled directory scanning and log generation for tracking and analysis.

EDUCATION

Graphic Era Deemed to be University, Dehradun, India 2023 – 2026

BCA with specialization in Artificial Intelligence & Data Science

CGPA: 6.8/10

CERTIFICATIONS & ACHIEVEMENTS

- Completed Advanced Machine Learning Course (Coursera), 2025
- Human Computer Interaction (Elite) – NPTEL, IIT Madras
- Deloitte Technology Job Simulation – Forage
- TATA GenAI Powered Data Analytics Job Simulation – Forage